

CHEATSHEET · QUICK REFERENCE

# Lab 22 — Auditing, Logging, and SIEM Basics

Write auditd watch rules, forward via rsyslog, tune rotation/retention, run a logwatch report and simulate the five-stage SIEM workflow with a correlation rule.

## auditd watch rules

```
-w /etc/passwd -p wa -k id_changes
Watch identity file changes

-w /etc/sudoers -p wa -k privesc
Watch privilege escalation config

-a always,exit -F arch=b64 -S unlink,rename -F dir=/etc -k file_delete
Watch deletions in /etc

augenrules --load
Load rules

ausearch -k id_changes -ts recent
Search matched events
```

## rsyslog severity levels

| Level | Name    |
|-------|---------|
| 0     | emerg   |
| 1     | alert   |
| 2     | crit    |
| 3     | err     |
| 4     | warning |
| 6     | info    |
| 7     | debug   |

## Rotation, report and forwarding

```
rotate 12 # keep ~90 days
Retention aligned to policy

logwatch --output stdout --range today --detail Med
Daily summary report

*. * @@siem.lab.local:6514
Forward via TCP (@@ = reliable)
```

## SIEM workflow + controls

| Stage / control      | Meaning                           |
|----------------------|-----------------------------------|
| Collect / Normalise  | Ship + parse to common schema     |
| Correlate / Alert    | Rule match (5 fails -> 1 success) |
| Investigate          | Pivot on user/host/time           |
| Two-person integrity | Critical ops need two people      |
| Separation of roles  | Request != approve != implement   |